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United States Patent Application Publication

20250275787

Kind Code

A1

Publication Date

September 04, 2025

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METHODS FOR ACCESSING NERVES WITHIN BONE

Abstract

Methods of ablating a basivertebral nerve (BVN) in a lumbar or sacral vertebral body of a patient. An introducer and one or more instruments are selected after a pre-surgical evaluation. The introducer has a handle coupled to a shaft defining a central channel and an opening at a distal tip thereof. The introducer is inserted through a pedicle of the vertebral body such that its opening reaches a cancellous portion of the vertebral body. The one or more instruments are delivered through the central channel. A distal deflectable section of the one or more instruments is advanced past the introducer's opening to position it proximate to the BVN in a posterior midline section of the vertebral body. An active element is extended relative to at least a portion of the one or more instruments. Energy is delivered to the active element to ablate at least a portion of the BVN.

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Family ID: 42060393

Appl. No.: 19/212313

Filed: May 19, 2025

Related U.S. Application Data

parent US continuation 18778387 20240719 parent-grant-document US 12303166 child US 19212313

parent US continuation 16368453 20190328 parent-grant-document US 12329412 child US 18778387

parent US continuation 15040268 20160210 parent-grant-document US 10265099 child US 16368453

parent US continuation 13862242 20130412 parent-grant-document US 9259241 child US 15040268

parent US continuation 12566895 20090925 parent-grant-document US 8419730 child US

Publication Classification

Int. Cl.: **A61B17/34** (20060101); **A61B17/16** (20060101); **A61B17/88** (20060101); **A61B18/00** (20060101); **A61B18/02** (20060101); **A61B18/14** (20060101); **A61B18/18** (20060101); **A61B18/20** (20060101); **A61N7/00** (20060101); **A61N7/02** (20060101)

U.S. Cl.:

CPC **A61B17/3472** (20130101); **A61B17/16** (20130101); **A61B17/1642** (20130101); **A61B17/1671** (20130101); **A61B17/34** (20130101); **A61B17/3403** (20130101); **A61B17/3421** (20130101); **A61B17/8819** (20130101); **A61B18/14** (20130101); **A61B18/148** (20130101); **A61B18/1487** (20130101); **A61B18/149** (20130101); **A61B18/18** (20130101); **A61B18/1815** (20130101); **A61B18/20** (20130101); **A61N7/00** (20130101); A61B2018/00011 (20130101); A61B2018/00017 (20130101); A61B2018/00339 (20130101); A61B2018/00434 (20130101); A61B2018/00577 (20130101); A61B18/0206 (20130101); A61B18/1492 (20130101); A61N2007/0026 (20130101); A61N2007/025 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent Ser. No. 18/778,387, filed on Jul. 19, 2024, which is a continuation of U.S. patent application Ser. No. 16/368,453, filed on Mar. 28, 2019, which is a continuation of U.S. patent application Ser. No. 15/040,268, filed on Feb. 10, 2016, issued as U.S. Pat. No. 10,265,099, which is a continuation of U.S. patent application Ser. No. 13/862,242 filed on Apr. 12, 2013, issued as U.S. Pat. No. 9,259,241, which is a continuation of U.S. patent application Ser. No. 12/566,895 filed on Sep. 25, 2009, issued as U.S. Pat. No. 8,419,730, which claims the benefit of U.S. Provisional Application 61/100,553 filed on Sep. 26, 2008, each of which is expressly incorporated in its entirety by reference herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] This invention pertains generally to generating passageways through tissue, and more particularly to creating curved paths in bone.

2. Description of Related Art

[0003] Recently, the technique of accessing the vertebral body through minimally invasive means has been developed through the surgical techniques used in vertebroplasty and kyphoplasty. Although accessing the vertebral segments of the spine through the pedicle and into the lateral/anterior section of the body of the vertebra is the primary method of placing a treatment device (e.g. a bone cement delivery device and/or an RF probe) into the vertebra, it is difficult to place a probe in the posterior midline section of the vertebra. Furthermore, accessing the posterior midline section of the S1 segment of the spine is difficult with a straight linear access route. A probe preferably needs to be capable of navigating to the posterior section of the S1 vertebral body as well as the same target area within a lumbar vertebral segment. In addition, it is contemplated that spinal segments in the cervical and thoracic spine may also be targeted.

[0004] In order to accurately and predictably place a treatment device in the posterior midline

section of a lumbar vertebral body or S1 vertebral body, the device or probe needs to navigate to said area through varying densities of bone. However, due to the varying densities of bone, it is difficult to navigate a probe in bone and ensure its positioning will be in the posterior midline section of the vertebral body.

[0005] Current techniques for tissue aspirations require a coaxial needle system that allows taking several aspirates through a guide needle without repositioning the guide needle. However, the problem with this system is that after the first pass of the inner needle in to the lesion, subsequent passes tend to follow the same path within the mass, yielding only blood not diagnostic cells.

[0006] A scientific paper written by Kopecky et al., entitled "Side-Exiting Coaxial Needle for Aspiration Biopsy," describes the use of a side exiting coaxial needle to allow for several aspiration biopsies. The guide needle has a side hole 1 cm from the distal tip. When a smaller needle is advanced through this new guide needle, the smaller needle is deflected by a ramp inside the guide, causing the smaller needle to exit through the side hole. Although this side exiting needle is able to deflect a bone aspiration needle, it does not guarantee that the needle exits the side hole in a linear direction into the tissue site. Once the tissue aspiration needle exits the needle, it will deviate from a linear path depending on the density of the tissue and inherent material strength of the needle. This is an inherent problem the device is unable to overcome.

[0007] Accordingly, an object of the present invention is a system and method for generating a path in bone that predictably follows a predetermined curved path.

BRIEF SUMMARY OF THE INVENTION

[0008] The present invention is directed to systems and methods to deploy and navigate a flexible treatment instrument, such as an RF bipolar probe, within bone. Although the systems and methods described below are primarily directed to navigating bone through a vertebral member of the spine, and particularly to treat the BVN of a vertebral member, it is appreciated that the novel aspects of the present invention may be applied to any tissue segment of the body.

[0009] The first novel principle of this invention is the ability to navigate a curve or angle within varying densities of cancellous bone and create a straight channel at the end of the navigated curve or angle. Several systems are described.

[0010] One aspect is a method of therapeutically treating a vertebral body having an outer cortical bone region and an inner cancellous bone region, and a BVN having a trunk extending from the outer cortical bone region into the inner cancellous region and a branches extending from the trunk to define a BVN junction, comprising the steps of: a) inserting an energy device into the vertebral body, and b) exclusively depositing energy within the inner cancellous bone region of the vertebral body between, but exclusive of the BVN junction and the outer cortical bone region, to denervate the BVN.

[0011] In another aspect of the present invention, a tube-within-tube embodiment has a deployable curved Nitinol tube that deploys from a straight cannula. The Nitinol tube is pre-curved to create an angular range of approximately 0° to approximately 180°, but more specifically from approximately 45° to approximately 110°, when fully deployed from the straight cannula. The design of the curve is such that the flexible element (carrying the treatment device) can navigate through the angular range of deployment of the nitinol tube. The curved nitinol tube allows the flexible element to navigate through a curve within bone without veering off towards an unintended direction. Cancellous bone density varies from person to person. Therefore, creating a curved channel within varying density cancellous bone will generally not predictably or accurately support and contain the treatment device as it tries to navigate the curved channel. With the present invention, the flexible element is deployed into the bone through the curved Nitinol tube, which supports the element as it traverses through the curve. When it departs from the tube, it will do so in a linear direction towards the target zone. This design allows the user to predictably and accurately deploy the flexible element towards the target zone regardless of the density of the cancellous bone.

[0012] An aspect of the invention is a system for channeling a path into bone. The system comprises a trocar having a central channel and opening at its distal tip, and a cannula sized to be received in said central channel and delivered to said distal opening. The cannula has a deflectable tip with a preformed curve such that the tip straightens while being delivered through the trocar and regains its preformed curve upon exiting and extending past the distal opening of the trocar to generate a curved path in the bone corresponding to the preformed curve of the deflectable tip. The cannula comprises a central passageway having a diameter configured allow a treatment device to be delivered through the central passageway to a location beyond the curved path.

[0013] In one embodiment, the system further includes a straight stylet configured to be installed in the trocar, wherein the straight stylet comprises a sharp distal tip that is configured to extend beyond the distal opening of the trocar to pierce the bone as the trocar is being delivered to a treatment location within the bone.

[0014] The system may further include a straightening stylet configured to be installed in the cannula, wherein the straightening stylet comprising a rigid construction configured to straighten the distal tip of the cannula when positioned in the trocar.

[0015] In an alternative embodiment, the straightening stylet further comprises a sharp distal end to pierce the bone, and the straightening stylet and cannula are installed in the trocar in place of the straight stylet as the trocar is delivered into the bone.

[0016] In a preferred embodiment, the system further includes a curved stylet having an outer radius sized to fit within the central passageway of the curved cannula. The curved stylet is configured to be installed in the curved cannula while the curved cannula is extended past the distal opening of the trocar, the curved stylet configured to block the distal opening of the curved cannula while being delivered into the bone. Preferably, the curved stylet has a curved distal end corresponding to the curve of the curved cannula.

[0017] The curved stylet also has a sharp distal tip configured to extend past the curved cannula to pierce the bone as the cannula is delivered past the distal opening of the trocar. The curved stylet also preferably comprises an angled distal tip configured to further support and maintain the curved stylet radius as it is delivered past the distal opening of the trocar and into bone.

[0018] Preferably, the curved stylet and the curved cannula have mating proximal ends that align the curve of the curved stylet with the curve of the curved cannula.

[0019] In one embodiment, the system further includes a straight channeling stylet configured to be installed in the cannula after removing the curved stylet, wherein the straight channeling stylet is flexibly deformable to navigate the curved cannula yet retain a straight form upon exiting the curve cannula, and wherein straight channeling stylet has a length longer than the curved cannula such that it creates a linear path beyond the distal end of the curved cannula when fully extended.

[0020] Another aspect is method for channeling a path into bone to a treatment location in the body of a patient. The method includes the steps of inserting a trocar having a central channel and opening at its distal tip into a region of bone at or near the treatment location, and delivering a cannula through said central channel and to said distal opening, wherein the cannula comprises a deflectable tip with a preformed curve such that the tip straightens while being delivered through the trocar and regains its preformed curve upon exiting the trocar, and extending the cannula past the distal opening of the trocar to generate a curved path in the bone corresponding to the preformed curve of the deflectable tip. Finally, a treatment device is delivered through a central passageway in said cannula to the treatment location beyond the curved path.

[0021] In one embodiment, inserting a trocar into a region of bone comprises inserting a stylet into the trocar such that the stylet extends beyond the distal opening of the trocar, and inserting the stylet and trocar simultaneously into the region of bone such that the stylet pierces the bone as the trocar is being delivered to a treatment location.

[0022] In another embodiment, delivering a cannula through the central channel comprises inserting a straightening stylet into the central passageway of the cannula, wherein the straightening

stylet comprises a rigid construction configured to straighten the curved distal tip of the cannula, and inserting the straightening stylet and straightened cannula simultaneously into the trocar.

[0023] In an alternative embodiment, the straightening stylet further comprises a sharp distal end to pierce the bone, wherein the straightening stylet and cannula are installed simultaneously along with the trocar as the trocar is delivered into the bone.

[0024] In yet another embodiment, extending the cannula past the distal opening is done by inserting a curved stylet into the central passageway of the curved cannula such that a distal tip of the curved stylet extends to at least the distal opening of the curved cannula, and simultaneously extending the curved cannula and curved stylet from the distal end of the trocar such that the curved stylet blocks the distal opening of the curved cannula while being delivered into the bone.

[0025] In a preferred embodiment, the curved stylet has a curved distal end corresponding to the curve of the curved cannula, and wherein the curved stylet reinforces the curved shape of the curved cannula as the curved cannula is extended past the distal opening of the trocar. The curved stylet has a sharp distal tip such that it is advanced within the central passageway so that the curved stylet extends past the distal opening of the curved cannula such that the curved stylet pierces the bone as the cannula is delivered past the distal opening of the trocar.

[0026] In a further step, the curved stylet is removed from the curved cannula, and a straight channeling stylet is inserted into the curved distal end of the cannula. The straight channeling stylet is flexibly deformable to navigate the curved cannula, yet retain a straight form upon exiting the curved cannula. The straight channeling stylet is longer than the curved cannula to create a linear channel beyond the distal tip of the curved cannula.

[0027] In a preferred embodiment, the trocar is inserted through a cortical bone region and into a cancellous bone region of a vertebrae, and the curved cannula is extended though at least a portion of the cancellous bone region to a location at or near the treatment location. A preferred treatment location comprises a BVN of the vertebrae, and treatment is delivered to the treatment location to denervate at least a portion of the BVN. In one embodiment, a portion of the BVN is denervated by delivering focused, therapeutic heating to an isolated region of the BVN. In another embodiment, a portion of the BVN comprises is denervated delivering an agent to the treatment region to isolate treatment to that region. Preferably, the treatment is focused on a location of the BVN that is downstream of one or more branches of the BVN.

[0028] Another aspect is a kit for channeling a path into bone. The kit includes a trocar having a central channel and opening at its distal tip, and a cannula selected from a set of cannulas sized to be received in said central channel and delivered to said distal opening. The cannula has a deflectable distal tip with a preformed curve such that the tip straightens while being delivered through the trocar and regains its preformed curve upon exiting and extending past the distal opening of the trocar to generate a curved path in the bone corresponding to the preformed curve of the deflectable tip. The cannula comprises a central passageway having a diameter configured allow a treatment device to be delivered through the central passageway to a location beyond the curved path, wherein the set of cannulas comprises one or more cannulas that have varying preformed curvatures at the distal tip.

[0029] In a preferred embodiment, the one or more cannulas have a varying preformed radius at the distal tip. In addition, the one or more cannulas each have distal tips that terminate at varying angles with respect to the central channel of the trocar. The length of the distal tips may also be varied. The angle of the distal with respect to the central channel of the trocar may vary from 0 degrees to 180 degrees.

[0030] The kit may further include a straight stylet configured to be installed in the trocar, the straight stylet comprising a sharp distal tip that is configured to extend beyond the distal opening of the trocar to pierce the bone as the trocar is being delivered to a treatment location within the bone.

[0031] In a preferred embodiment, the kit includes a set of curved stylets having an outer radius sized to fit within the central passageway of the curved cannula, wherein each curved stylet is

configured to be installed in the curved cannula while the curved cannula is extended past the distal opening of the trocar. The curved stylet is configured to block the distal opening of the curved cannula while being delivered into the bone. Each curved stylet has a varying curved distal end corresponding to the curve of a matching curved cannula in the set of curved cannulas. The curved stylet has a sharp distal tip configured to extend past the curved cannula to pierce the bone as the cannula is delivered past the distal opening of the trocar.

[0032] In another embodiment, the kit includes a set of straight channeling stylets wherein one of the set of stylets is configured to be installed in the cannula after removing the curved stylet. The straight channeling stylet is flexibly deformable to navigate the curved cannula yet retain a straight form upon exiting the curve cannula. Each of the straight channeling stylets has a varying length longer than the curved cannula such that the straight channeling stylet creates a predetermined-length linear path beyond the distal end of the curved cannula when fully extended.

[0033] Further aspects of the invention will be brought out in the following portions of the specification, wherein the detailed description is for the purpose of fully disclosing preferred embodiments of the invention without placing limitations thereon.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0034] The invention will be more fully understood by reference to the following drawings which are for illustrative purposes only:

[0035] FIG. 1 is a system for generating a curved path in bone according to the present invention.

[0036] FIG. 2 is a sectional view of the system of FIG. 1

[0037] FIG. 3 illustrates a sectioned view of a vertebral body with a path bored through the cortical shell.

[0038] FIGS. 4A-F illustrate a method for accessing the BVN with the system of the present invention.

[0039] FIG. 5 shows an alternative system for generating a curved path in bone according to the present invention.

[0040] FIG. 6 shows the system of FIG. 5 being installed in a vertebral body.

[0041] FIGS. 7A-7B show a curved stylet in accordance with the present invention.

[0042] FIG. 8 shows an embodiment of a windowed cannula kit.

[0043] FIG. 9 shows an embodiment of a reverse ramp window stylet disposed within a windowed needle of the windowed cannula kit of FIG. 8.

[0044] FIG. 10 shows an embodiment of a channeling stylet extending from a window defined in the windowed needle of the windowed cannula kit of FIG. 8.

[0045] FIG. 11 shows a flexible functional element with an RF probe at its distal tip extending from the window defined in the windowed needle of the windowed cannula kit of FIG. 8.

[0046] FIG. 12 is a detailed view of the distal end of the windowed needle of the windowed cannula kit of FIG. 8.

DETAILED DESCRIPTION OF THE INVENTION

[0047] Referring more specifically to the drawings, for illustrative purposes the present invention is embodied in the apparatus generally shown in FIG. 1 through FIG. 7B. It will be appreciated that the apparatus may vary as to configuration and as to details of the parts, and that the method may vary as to the specific steps and sequence, without departing from the basic concepts as disclosed herein.

[0048] FIGS. 1 and 2 illustrate a first embodiment of the present invention comprising a system or kit 10 for forming a path through bone. The system comprises a needle trocar 20 (the main body of the instrument set). The trocar 20 comprises an elongate shaft 28 having a handle 24 at its proximal

end **32** and a central lumen **36** passing through to the distal end **22** of the trocar **20**. The central lumen **36** is generally sized to allow the other instruments in the system **10** to be slideably introduced into the patient to a treatment region. System **10** further comprises a straight stylet **80** having a sharp-tipped needle **84** at its distal end that is used with the needle trocar **20** to create the initial path through the soft tissue and cortical shell to allow access to the cancellous bone, a curved cannula **50** that is used to create/maintain the curved path within the bone/tissue. A straightening stylet **40** is used to straighten out the curve and load the curved cannula **50** into the needle trocar **20**. A curved stylet **60** is used in conjunction with the curved cannula **50** to create the curved path within the bone/tissue, and a channeling stylet **90** is used to create a working channel for a treatment device (such as RF probe **100**) beyond the end of the curved path created by the curved cannula **50**.

[0049] The surgical devices and surgical systems described may be used to deliver numerous types of treatment devices to varying regions of the body. Although the devices and systems of the present invention are particularly useful in navigating through bone, it is appreciated that they may also be used to navigate through soft tissue, or through channels or lumens in the body, particularly where one lumen may branch from another lumen.

[0050] The following examples illustrate the system **10** applied to generating a curve bone path in the vertebral body, and more particularly for creating a bone path via a transpedicular approach to access targeted regions in the spine. In particular, the system **10** may be used to deliver a treatment device to treat or ablate intraosseous nerves, and in particular that basivertebral nerve (BVN). Although the system and methods provide significant benefit in accessing the BVN, it is appreciated that the system **10** of the present invention may similarly be used to create a bone path in any part of the body.

[0051] FIG. **3** illustrates a cross-sectional view of a vertebra **120**. Recently, the existence of substantial intraosseous nerves **122** and nerve branches **130** within human vertebral bodies (“basivertebral nerves”) have been identified. The basivertebral nerve **122** has at least one exit **142** point at a location along the nerve **122** where the nerve **122** exits the vertebral body **126** into the vertebral foramen **132**.

[0052] Preferably, the basivertebral nerves are at, or in close proximity to, the exit point **142**. Thus, the target region of the BVN **122** is located within the cancellous portion **124** of the bone (i.e., to the interior of the outer cortical bone region **128**), and proximal to the junction J of the BVN **122** having a plurality of branches **130** (e.g. between points A and B along nerve **122**). Treatment in this region is advantageous because only a single portion of the BVN **122** need be effectively treated to denervate or affect the entire system. Typically, treatment in accordance with this embodiment can be effectuated by focusing in the region of the vertebral body located between 60% (point A) and 90% (point B) of the distance between the anterior and posterior ends of the vertebral body. In contrast, treatment of the BVN **122** in locations more downstream than the junction J require the denervation of each branch **130**.

[0053] In one approach for accessing the BVN, the patient's skin is penetrated with a surgical instrument which is then used to access the desired basivertebral nerves, i.e., percutaneously. In one embodiment, a transpedicular approach is used for penetrating the vertebral cortex to access the BVN **122**. A passageway **140** is created between the transverse process **134** and spinous process **136** through the pedicle **138** into the cancellous bone region **124** of the vertebral body **126** to access a region at or near the base of the nerve **122**. It is appreciated that a posterolateral approach (not shown) may also be used for accessing the nerve.

[0054] FIGS. **4A-F** illustrate a preferred method for accessing the BVN with the system **10** of the present invention. First, the straight stylet **80** is inserted in aperture **26** at the proximal end **32** of needle trocar **20**. The straight stylet **80** is advanced down the central lumen **36** (see FIG. **2**) of the trocar **20** until the proximal stop **82** abuts against handle **24** of the trocar **20**, at which point the distal tip **84** of straight stylet protrudes out of the distal end **22** of the trocar **20**. The tip **84** of the

straight stylet **80** preferably comprises a sharp tip for piercing soft tissue and bone.

[0055] Referring now to FIG. **4A**, the assembly (trocar **20** and straight stylus **80**) is advanced through soft tissue to the surface of the bone. Once the proper alignment is determined, the assembly is advanced through the cortical shell of pedicle **138** and into the cancellous interior **124** of the bone.

[0056] After the proper depth is achieved, the straight stylet **80** is removed from the trocar **20**, while the trocar **20** remains stationary within the vertebrae **120**. The straightening stylet **40** is inserted into proximal aperture **53** (see FIG. **2**) of the curved cannula **50** and advanced along the central lumen of the curved cannula **50** until the stop **42** of the stylet **40** abuts up to the proximal end **52** of the curved cannula. This forces the distal tip of the straight stylet through the curved section **56** of the curved cannula **50** to straighten out the curved section **56**. It is contemplated that the straight stylet comprise a hard, non-compliant material and the distal end of the curved cannula **50** a compliant, yet memory retaining material (e.g. Nitinol, formed PEEK, etc.) such that the curved section **56** yields to the rigidity of the straightening stylet **40** when installed, yet retains its original curved shape when the stylet **40** is removed.

[0057] As shown in FIG. **4B**, once the straightening stylet **40** is secure and the curved cannula **50** is straight, they are inserted into the needle trocar **20** and secured. Proper alignment (e.g. prevent rotation, orient curve direction during deployment) is maintained by aligning a flat on the upper portion **58** of the curved cannula **50** to an alignment pin secured perpendicularly into the needle trocar **20** handle **24**. Once the curved cannula **50** is secure, the straightening stylet **40** is removed, while the curved cannula **50** remains stationary within the trocar **20**.

[0058] Referring to FIG. **4C**, the curved stylet **60** is then straightened out by sliding the small tube **68** proximally to distally on its shaft towards the distal tip **64** or from the distal tip **64** proximally on its shaft towards the proximal end **62**. Once the curved distal tip **66** is straightened out and fully retracted inside the small tube **68**, the curved stylet **60** is inserted into the proximal aperture **53** of the curved cannula **50**, which still resides inside the needle trocar **20**. As the curved stylet **60** is advanced into the curved cannula **50**, the small tube **68** is met by a stop **55** (see FIG. **4C**). As the curved stylet **60** continues to advance the small tube **68** is held inside the handle of the curved cannula **50**. This allows the curve of the stylet **60** to be exposed inside the curved cannula **50**. To create the maximum force the curve of the two parts (**50** & **60**) must be aligned. To ensure alignment the cap on the curved stylet **60** has an alignment pin **70** which engages with an alignment notch on the proximal end **52** of the curved cannula **50**.

[0059] Once the stylet **60** is fully seated and aligned with the curved cannula **50** the tip of the curved stylet **60** will protrude from the tip of the curved cannula **50** by about 1/16 to 3/16 inches. This protrusion will help to drive the curve in the direction of its orientation during deployment.

[0060] Referring now to FIG. **4D**, with the curved stylet **60** and the curved cannula **50** engaged, the locking nut **58** at the top of the curved cannula **50** is rotated counter clockwise to allow the cannula **50** and curved stylet **60** to be advanced with relation to the needle trocar **20** such that the proximal end **52** abuts against **58**, advancing the curved cannula **50** and curved stylet **60** beyond the distal opening of trocar **20** to generate a curved path in the cancellous bone region **124**. As the curved cannula **50** and curved stylet **60** are advanced they will preferably curve at a radius of 0.4 to 1.0 inches through cancellous bone and arc to an angle between 5 and 110 degrees. Once the curved cannula **50** and curved stylet **60** are deployed to the intended angle, the locking nut at the top of the curved cannula **50** is engaged with the needle trocar **20** to stop any additional advancement of the curved stylet cannula assembly.

[0061] FIGS. **7A-7B** illustrate the tip of the curved stylet **60**, which has been formed with two angles. To help the curve deployment in the proper direction the curved section **66** of the curved stylet **60** is shaped in a predetermined orientation. The angle on the inside of the curve **72** is less than the angle on the outside of the curve **74**. This disparity in angle helps the stylet cannula assembly **50** & **60** curve in the bone as bone pushes against outside curve face **74** ensuring the

curve radius is maintained during deployment.

[0062] Referring now to FIG. 4E, the curved stylet **60** is then removed and replaced by the channeling stylet **90**. The tip **94** of the channeling stylet **90** is advanced beyond the end **54** of the curved cannula **50** towards the intended target treatment zone.

[0063] Referring now to FIG. 4F, once the channeling stylet **90** reaches the target treatment zone, it is removed creating a working channel **146**. Channel **140** will generally have a first section **142** that crosses the cortical bone of the pedicle **138**, followed by a curved path **144**. These sections are occupied by curved cannula **50** such that a treatment device fed through the cannula **50** will have to follow the curve of the cannula **50** and not veer off in another direction. The channel may further comprise the linear extension **146** in the cancellous bone **124** to further advance the treatment device toward the treatment site T.

[0064] With the trocar **20** and curved cannula **50** still in place, a treatment device (e.g. treatment probe **100** shown in FIG. 2, with an active element **102** on the distal end **104** of elongate flexible catheter **110** is delivered to the target treatment location T to perform a localized treatment.

[0065] In a preferred embodiment, the active element **102** is delivered to the treatment site and activated to delivery therapeutic treatment energy. The treatment probe may comprise an RF delivery probe having bipolar electrodes **106** and **108** that deliver a therapeutic level of heating to stimulate or ablate the nerve **122**.

[0066] It is appreciated that any number of treatment modalities may be delivered to the treatment site for therapeutic treatment. For example, treatment may be affected by monopolar or tripolar RF, ultrasound, radiation, steam, microwave, laser, or other heating means. Additionally, the treatment device may comprise a fluid delivery catheter that deposits an agent, e.g. bone cement, or other therapeutic agent, to the treatment site T. Alternatively, cryogenic cooling may be delivered for localized treatment of the BVN. Furthermore, treatment may be affected by any mechanical destruction and or removal means capable of severing or denervating the BVN. For example, a cutting blade, bur or mechanically actuated cutter typically used in the art of orthoscopic surgery may be used to affect denervation of the BVN.

[0067] In addition to or separate from treating the BVN, a sensor may be delivered to the region to preoperatively or postoperatively measure nerve conduction at the treatment region. In this configuration, the sensor may be delivered on a distal tip of a flexible probe that may or may not have treatment elements as well.

[0068] The goal of the treatment may be ablation, or necrosis of the target nerve or tissue, or some lesser degree of treatment to denervate the BVN. For example, the treatment energy or frequency may be just sufficient to stimulate the nerve to block the nerve from transmitting signal (e.g. signals indicating pain).

[0069] Once the treatment is complete, the probe **100** is withdrawn. The curved cannula **50** is then withdrawn into the needle trocar **20**. The needle trocar **20** with the curved cannula **50** is then removed and the access site is closed as prescribed by the physician.

[0070] In the above system **10**, the design of the curved sections **56** and **66** of the curved cannula **50** and curved stylet **60**, respectively, is such that the flexible element (e.g. carrying the treatment device) can navigate through the angular range of deployment of the Nitinol tube of the curved cannula **50**. The curved nitinol tube of the curved cannula **50** allows the flexible element to navigate through a curve within bone without veering off towards an unintended direction. Cancellous bone density varies from person to person. Therefore, creating a curved channel within varying density cancellous bone **124** will generally not predictably or accurately support and contain the treatment device as it tries to navigate the curved channel.

[0071] With the system **10** of the present invention, the treatment device **100** is deployed into the bone through the curved Nitinol tube of the curved cannula **50**, which supports the element as it traverses through the curve. When it departs from the tube, it will do so in a linear direction along path **146** towards the target zone. This allows the user to predictably and accurately deploy the

treatment device towards the target zone T regardless of the density of the cancellous bone.

[0072] In some embodiments, a radius of curvature that is smaller than that which can be achieved with a large diameter Nitinol tube may be advantageous. To achieve this, the curved tube of the curved cannula **50** may take one of several forms. In one embodiment, the tube **50** is formed from a rigid polymer that can be heat set in a particular curve. If the polymer was unable to hold the desired curve, an additional stylet (e.g. curved stylet **60**) of Nitinol, or other appropriate material, may also be used in conjunction with the polymer tube to achieve the desired curve. This proposed combination of material may encompass and number or variety of materials in multiple different diameters to achieve the desired curve. These combinations only need to ensure that the final outside element (e.g. trocar **20**) be “disengageable” from the internal elements and have an inner diameter sufficient to allow the desired treatment device **100** to pass to the treatment region T.

[0073] In an alternative embodiment, of the curved cannula **50** may comprise a Nitinol tube having a pattern of reliefs or cuts (not shown) in the wall of the tube (particularly on the outer radius of the bend). The pattern of cuts or reliefs would allow the tube to bend into a radius tighter than a solid tube could without compromising the integrity of the tubing wall.

[0074] FIG. 5 illustrates a second embodiment of the system or kit **200** of the present invention that may be used to reduce the number of steps required for the procedure. The second embodiment includes a needle trocar **20**, straightening stylet **40**, used with the needle trocar **20** and the curved cannula **50** to create the initial path through the soft tissue and cortical shell to allow access to the cancellous bone, curved stylet **60** used in conjunction with the curved cannula **50** to create the curved path within the bone/tissue, and channeling stylet **90** used to create a working channel for the probe beyond the end of the curved path created by the curved stylet.

[0075] In one method according to the present invention, the straightening stylet **40** is inserted into the curved cannula **50** and secured. In this embodiment, the straightening stylet **40** has a sharp tip **46** designed to penetrate bone. Once the straightening stylet **40** is secure and the curved cannula **50** is straight, they are inserted into the needle trocar **20** and secured. In this embodiment, the curved cannula **50** and straightening stylet **40** are inserted into the shaft **28** of the trocar **20** only as far as to have sharp tip **46** of the straightening stylet **40** protrude from the distal end **22** of the trocar **20**. Proper alignment is maintained by aligning a flat on the upper portion of the curved cannula **50** with a pin secured perpendicularly into the needle trocar **20** handle.

[0076] Referring now to FIG. 6, once the curved cannula **50** is secure, the assembly (trocar **20**, curved cannula **50**, and straightening stylet **40**) is advanced through soft tissue to the surface of the bone. After finding the proper alignment at the pedicle **138** of vertebrae **120**, the assembly (trocar **20**, curved cannula **50**, and straightening stylet **40**) is advanced through the cortical shell **128** and into the cancellous interior **124** of the bone.

[0077] After the proper depth is achieved, the straightening stylet **40** is removed. The curved stylet **60** is then straightened out by sliding the small tube **68** on its shaft towards the distal tip **64**. The curved distal tip **66** is straightened out and fully retracted inside the small tube **68**, and then the curved stylet **60** is inserted into the curved cannula **50** which still resides inside the needle trocar **20**. Once the curved stylet **60** is inserted into the curved cannula **50**, the small tube **68** is met by a stop **55** (see FIG. 4C). As the curved stylet **60** continues to advance, the small tube **68** is held inside the handle of the curved cannula **50**. This allows the curve of the stylet **60** to be exposed inside the curved cannula **50**.

[0078] To create the maximum force, it is preferred that the curve of the two parts (**50** & **60**) are aligned. To ensure alignment the cap on the curved stylet **60** has an alignment pin, which engages with a notch on the top of the curved cannula **50**.

[0079] When the stylet **60** is fully seated and aligned with the curved cannula **50**, the tip of the curved stylet **60** will protrude from the tip of the curved cannula **50** by about 1/16 to 3/16 inches. This protrusion will help to drive the curved cannula **50** in the direction of its orientation during deployment. Once the curved stylet **60** and the curved cannula **50** are engaged, the lock nut at the

top of the curved cannula **50** is rotated counter clockwise to allow the cannula **50** and stylet **60** to be advanced with relation to the needle trocar **20** (as shown in FIG. 4D). As the curved cannula and stylet are advanced they generate a curved path toward the treatment location T. Once the curved cannula **50** and stylet **60** are deployed to the intended angle, the lock nut at the top of the curved cannula **50** is engaged with the needle trocar **20** to stop any additional advancement of the curved stylet cannula assembly.

[0080] The curved stylet **60** is then removed and replaced by the channeling stylet **90**. The channeling stylet **90** is advanced beyond the end of the curved cannula **50** (see FIG. 4E) towards the intended target treatment zone creating a working channel for the active element to be inserted. Once the channeling stylet **80** reached the target treatment zone it is removed and replaced by the treatment device **100**, which is delivered to the treatment site T and activated.

[0081] Once the treatment is complete, the treatment device **100** is withdrawn. The curved cannula **50** is then withdrawn into the needle trocar **20**. The needle trocar **20** with the curved cannula **50** is then removed and the access site is closed as prescribed by the physician.

[0082] FIGS. 7A and 7B illustrate detail views of a Nitinol wire for the curved stylet **60** (proximal end not shown). The wire comprises a shaft **78** having constant diameter D and a length L.sub.s that may vary according to the application and desired depth to the treatment location. The wire has a preformed distal tip that is curved to have a radius r that redirects the distal tip **64** at an angle Θ with the shaft. As shown in FIG. 7A, angle Θ is shown to be approximately 110°. However, it is appreciated that the preformed tip may have an angle ranging from a few degrees (slight deflection off axis), to up to 180° (e.g. directing back toward the proximal end).

[0083] As shown in FIG. 7B detailing the distal tip **64**, the tip may have a distal extension L.sub.T that extends away from the shaft **78**. To promote channeling along a path that follows radius r, the distal tip **64** is configured with a dual-plane bevels **74** and **72**. Plane **74** is offset at angle β , and plane **72** is offset at angle α . This configuration of the leading-allows for the stylet and/or curved cannula to travel through bone in a path correlating to the specified curve in the stylet and/or cannula.

[0084] In the example illustrated in FIGS. 7A and 7B, the curved stylet **60** has a shaft length L.sub.s of approximately 3.6 in., diameter D of approximately 0.040 in., and a distal tip length L.sub.T of 0.125 in., radius r of 0.40 in., and angle $\beta=35^\circ$ and angle $\alpha=31^\circ$. It should be noted that the above dimensions are for illustration only, and may vary depending on the anatomy an tissue type.

[0085] It is appreciated that all the above embodiments may be provided as a kit of instruments to treat different regions of the body. For example, the location, orientation and angle of the treatment device with respect to the trocar **20** may be varied by providing a set of instruments at varying increments. This may be achieved by varying the curvature (**56**, **66**) in the curved cannula **50** and curved stylet **60**. The curvature may be varied by varying the radius of curvature r, the insertion depth (shaft length L.sub.s and tip length L.sub.T, and/or the final exit angle Θ with respect to the trocar **20** central bore. Thus, the physician may select a different kit for treating a lumbar spine segment as opposed to a cervical spine segment, as the anatomy will dictate the path that needs to be channeled.

[0086] Thus, when treating different spine segments, a set out of the kit may be selected to match the vertebra (or other region being treated). For example, delivering the treatment device at or near the BVN junction for a lumbar vertebra may have a different angle than for a cervical vertebra, and may vary from patient to patient. The set may be selected from the kit intra-operatively, or from a pre-surgery diagnostic evaluation (e.g. radiographic imaging of the target region).

[0087] It is appreciated that each of the instruments in the embodiments 100 and 200 detailed above may have any length, shape, or diameter desired or required to provide access to the treatment region (e.g. intraosseous nerve trunk) thereby facilitating effective treatment of the target region. For example, the size of the intraosseous nerve to be treated, the size of the passageway in

the bone (e.g. pedicle **138**) for accessing the intraosseous nerve, and the location of the bone, and thus the intraosseous nerve, are factors that that may assist in determining the desired size and shape of the individual instruments.

[0088] The systems **100**, **200** described above may be used with a number of different treatment modalities for therapeutic treatment of the target region. For example, in one embodiment, it is desirable to operate the treatment device **100** in a manner that ablates the tissue of the target region (e.g. BVN) to produce as described in U.S. Pat. No. 6,699,242, herein incorporated by reference in its entirety.

[0089] In another embodiment, the treatment device is configured to deliver therapeutic treatment that is targeted to block nerve conduction without ablating the nerve, i.e. thermal treatment is delivered to the nerve (e.g. via thermal therapy, agent or the like) that results in denervation of the BVN without necrosis of tissue. This may be achieved via delivery of a lesser amount of energy or agent to the tissue site (either in the form of less exposure time, concentration, intensity, etc.) than is required for ablation, but an amount sufficient to achieve some amount of temporary or permanent denervation.

[0090] As can be seen, therefore, the present invention includes the following inventive embodiments among others: [0091] 1. A system for channeling a path into bone, comprising: a trocar having a central channel and opening at its distal tip; and a curved cannula sized to be received in said central channel and delivered to said distal opening; the curved cannula having a deflectable tip with a preformed curve such that the tip straightens while being delivered through the trocar and regains its preformed curve upon exiting and extending past the distal opening of the trocar to generate a curved path in the bone corresponding to the preformed curve of the deflectable tip; wherein the curved cannula comprises a central passageway having a diameter configured allow a treatment device to be delivered through the central passageway to a location beyond the curved path. [0092] 2. A system as recited in embodiment 2, further comprising: a straight stylet configured to be installed in the trocar; the straight stylet comprising a sharp distal tip that is configured to extend beyond the distal opening of the trocar to pierce the bone as the trocar is being delivered to a treatment location within the bone. [0093] 3. A system as recited in embodiment 2, further comprising: a straightening stylet configured to be installed in the curved cannula; the straightening stylet comprising a rigid construction configured to straighten the distal tip of the curved cannula when positioned in the trocar. [0094] 4. A system as recited in embodiment 3, wherein the straightening stylet further comprises a sharp distal end to pierce the bone; and wherein the straightening stylet and curved cannula are configured to be installed in the trocar in place of the straight stylet as the trocar is delivered into the bone. [0095] 5. A system as recited in embodiment 1, further comprising: a curved stylet having an outer radius sized to fit within the central passageway of the curved cannula; wherein the curved stylet is configured to be installed in the curved cannula while the curved cannula is extended past the distal opening of the trocar, the curved stylet configured to block the distal opening of the curved cannula while being delivered into the bone. [0096] 6. A system as recited in embodiment 5, wherein the curved stylet has a curved distal end corresponding to the curve of the curved cannula. [0097] 7. A system as recited in embodiment 5, wherein the curved stylet has a sharp distal tip configured to extend past the curved cannula to pierce the bone as the cannula is delivered past the distal opening of the trocar. [0098] 8. A system as recited in embodiment 7, wherein the curved stylet and the curved cannula have mating proximal ends that align the curve of the curved stylet with the curve of the curved cannula. [0099] 9. A system as recited in embodiment 5, further comprising a straight channeling stylet configured to be installed in the cannula after removing the curved stylet; wherein the straight channeling stylet is flexibly deformable to navigate the curved cannula yet retain a straight form upon exiting the curve cannula; wherein straight channeling stylet has a length longer than the curved cannula such that it creates a linear path beyond the distal end of the curved cannula when fully extended. [0100] 10. A method for channeling a path into bone to a treatment location in the

body of a patient, comprising: inserting a trocar into a region of bone near the treatment location; the trocar having a central channel and opening at its distal tip; and delivering a cannula through said central channel and to said distal opening; wherein the cannula comprises a deflectable tip with a preformed curve such that the tip straightens while being delivered through the trocar and regains its preformed curve upon exiting the trocar; extending the cannula past the distal opening of the trocar to generate a curved path in the bone corresponding to the preformed curve of the deflectable tip; delivering a treatment device through a central passageway in said cannula having to the treatment location beyond the curved path. [0101] 11. A method as recited in embodiment 10, wherein inserting a trocar into a region of bone comprises inserting a stylet into the trocar such that the stylet extends beyond the distal opening of the trocar; and inserting the stylet and trocar simultaneously into the region of bone such that the stylet pierces the bone as the trocar is being delivered to a treatment location. [0102] 12. A method as recited in embodiment 10, wherein delivering a cannula through the central channel comprises: inserting a straightening stylet into the central passageway of the cannula; the straightening stylet comprising a rigid construction configured to straighten the curved distal tip of the cannula; and inserting the straightening stylet and straightened cannula simultaneously into the trocar. [0103] 13. A method as recited in embodiment 12, wherein the straightening stylet further comprises a sharp distal end to pierce the bone; and wherein the straightening stylet and cannula are installed simultaneously along with the trocar as the trocar is delivered into the bone. [0104] 14. A method as recited in embodiment 10, wherein extending the cannula past the distal opening comprises: inserting a curved stylet into the central passageway of the curved cannula such that a distal tip of the curved stylet extends to at least the distal opening of the curved cannula; simultaneously extending the curved cannula and curved stylet from the distal end of the trocar such that the curved stylet blocks the distal opening of the curved cannula while being delivered into the bone. [0105] 15. A method as recited in embodiment 14, wherein the curved stylet has a curved distal end corresponding to the curve of the curved cannula; and wherein the curved stylet reinforces the curved shape of the curved cannula as the curved cannula is extended past the distal opening of the trocar. [0106] 16. A method as recited in embodiment 14, wherein the curved stylet has a sharp distal tip; wherein curved stylet is advanced within the central passageway so that the curved stylet extends past the distal opening of the curved cannula such that the curved stylet pierces the bone as the cannula is delivered past the distal opening of the trocar. [0107] 17. A method as recited in embodiment 14, further comprising: removing the curved stylet from the curved cannula; inserting a straight channeling stylet into the curved distal end of the cannula; wherein the straight channeling stylet is flexibly deformable to navigate the curved cannula yet retain a straight form upon exiting the curved cannula; wherein the straight channeling stylet is longer than the curved cannula to create a linear channel beyond the distal tip of the curved cannula. [0108] 18. A method as recited in embodiment 10, wherein the trocar is inserted through a cortical bone region and into a cancellous bone region of a vertebrae, and wherein the curved cannula is extended though at least a portion of the cancellous bone region to a location at or near the treatment location. [0109] 19. A method as recited in embodiment 18, wherein the treatment location comprises a BVN of the vertebrae, the method further comprising: delivering treatment to the treatment location to denervate at least a portion of the BVN. [0110] 20. A method as recited in embodiment 19, wherein denervating a portion of the BVN comprises delivering focused, therapeutic heating to an isolated region of the BVN. [0111] 21. A method as recited in embodiment 19, wherein denervating a portion of the BVN comprises delivering an agent to the treatment region to isolate treatment to that region. [0112] 22. A method as recited in embodiment 19, wherein the treatment is focused on a location of the BVN that is downstream of one or more branches of the BVN. [0113] 23. A kit for channeling a path into bone, comprising: a trocar having a central channel and opening at its distal tip; and a cannula selected from a set of cannulas sized to be received in said central channel and delivered to said distal opening; the cannula having a deflectable distal tip with a preformed curve such that the tip straightens while

being delivered through the trocar and regains its preformed curve upon exiting and extending past the distal opening of the trocar to generate a curved path in the bone corresponding to the preformed curve of the deflectable tip; wherein the cannula comprises a central passageway having a diameter configured allow a treatment device to be delivered through the central passageway to a location beyond the curved path; wherein the set of cannulas comprises one or more cannulas that have varying preformed curvatures at the distal tip. [0114] 24. A kit as recited in embodiment 23, wherein the one or more cannulas have a varying preformed radius at the distal tip. [0115] 25. A kit as recited in embodiment 23, wherein the one or more cannulas each have distal tips that terminate at varying angles with respect to the central channel of the trocar. [0116] 26. A kit as recited in embodiment 25, wherein the angle of the distal tip with respect to the central channel of the trocar varies from approximately 45° to approximately 110°. [0117] 27. A kit as recited in embodiment 23, further comprising: a straight stylet configured to be installed in the trocar; the straight stylet comprising a sharp distal tip that is configured to extend beyond the distal opening of the trocar to pierce the bone as the trocar is being delivered to a treatment location within the bone. [0118] 28. A kit as recited in embodiment 23, further comprising: a set of curved stylets having an outer radius sized to fit within the central passageway of the curved cannula; wherein each curved stylet is configured to be installed in the curved cannula while the curved cannula is extended past the distal opening of the trocar; wherein the curved stylet configured to block the distal opening of the curved cannula while being delivered into the bone; and wherein each curved stylet has a varying a curved distal end corresponding to the curve of a matching curved cannula in the set of curved cannulas. [0119] 29. A kit as recited in embodiment 28, wherein the curved stylet has a sharp distal tip configured to extend past the curved cannula to pierce the bone as the cannula is delivered past the distal opening of the trocar. [0120] 30. A kit as recited in embodiment 28, wherein the curved stylet and the curved cannula have mating proximal ends that align the curve of the curved stylet with the curve of the curved cannula. [0121] 31. A kit as recited in embodiment 28, further comprising a set of straight channeling stylets; wherein one of the set of stylets is configured to be installed in the cannula after removing the curved stylet; wherein the straight channeling stylet is flexibly deformable to navigate the curved cannula yet retain a straight form upon exiting the curve cannula; wherein each of the straight channeling stylets has a varying length longer than the curved cannula such that the straight channeling stylet creates a predetermined-length linear path beyond the distal end of the curved cannula when fully extended.

[0122] In accordance with yet another example, individual components of one embodiment of a system or kit are described in Table 1 below.

TABLE-US-00001 TABLE 1 Use Interfaces Identifier Description of the Curved Trocar “Tube in a Tube” System order with 1. Needle Trocar; the main body of the curved introducer 1 All except #2 instrument set. Used in conjunction with item #5 to create the path through the soft tissue and the cortical shell of the bone to allow access to the cancellous portion of the bone. 2. Straightening Stylet; used to straighten out the curve and load 2 Only # 3 Item # 3 into Item #1. After the soft tissue and cortical shell has been traversed by Items #1 and #5 in combination, item #5 is removed and Items #2 and #3 are inserted. Once Item # 3 is fully seated, Item # 2 is removed. 3. Curved Cannula: used to create/maintain the curved path within 3 All the bone/tissue. Used in conjunction with Item # 4 to create a curved path in the bone/tissue beyond the distal tip of Item #1. It allows item #4 to be withdrawn while maintaining the opening for the flexible probe or other flexible instrumentation. 4. Curved Stylet: Used in conjunction with Item # 3 to create the 4 Only #3 curved path within the Bone/Tissue. The straightening tube (4A) is used to straighten the tip of Item #4. Once the tip is in the straightened position Item #4/4A are inserted into Item #3. Once fully seated, and the curves are aligned, they are advanced together out of distal tip of Item # 1 and into the bone/tissue. The preset curve combination of Items #3 & #4 create a curved path through the bone/tissue. 5. Straight Stylet: used with Item#1 to create the initial path 1 Only # 1 through the soft tissue and cortical shell to allow access to the cancellous bone. 6. Channeling Stylet: Used to

create a working channel for the 5 Only #3 probe beyond the end of the curved path created by Items #3 & #4. Once the desired trajectory of the curved path has been achieved by items #3 & #4, item # 4 is removed. Item # 6 is inserted into Item # 3 and advanced through the curved path. Since Item # 6 is longer than Item # 3 it extends straight beyond the distal end of Item # 3 creating a straight working channel beyond the end of Item # 3

[0123] According to yet another example, individual components of one embodiment of a system or kit are described in Table 2 below.

TABLE-US-00002 TABLE 2 Use Interfaces Identifier Description of the Curved Trocar “Tube in a Tube” System order with 1. Needle Trocar; the main body of the curved introducer 1 All except #2 instrument set. Used in conjunction with item #5 to create the path through the soft tissue and the cortical shell of the bone to allow access to the cancellous portion of the bone. 2. Straightening Stylet; used to straighten out the curve and load 2 Only # 3 Item # 3 into Item #1 and then to create a path through the soft tissue and the cortical shell. After the soft tissue and cortical shell has been traversed by Items #1, #2 and # 3 in combination, item #2 is removed 3. Curved Cannula: used to create/maintain the curved path within 3 All the bone/tissue. Used in conjunction with Item # 4 to create a curved path in the bone/tissue beyond the distal tip of Item #1. It allows item #4 to be withdrawn while maintaining the opening for the flexible probe or other flexible instrumentation. 4. Curved Stylet: Used in conjunction with Item # 3 to create the 4 Only #3 curved path within the Bone/Tissue. The straightening tube (4A) is used to straighten the tip of Item #4. Once the tip is in the straightened position Item #4/4A are inserted into Item #3. Once fully seated, and the curves are aligned, they are advanced together out of distal tip of Item# 1 and into the bone/tissue. The preset curve combination of Items #3 & #4 create a curved path through the bone/tissue. 5. Channeling Stylet: Used to create a working channel for the 5 Only #3 probe beyond the end of the curved path created by Items #3 & #4 Once the desired trajectory of the curved path has been achieved by items #3 & #4, item # 4 is removed. Item # 5 is inserted into Item # 3 and advanced through the curved path. Since Item # 5 is longer than Item # 3 it extends straight beyond the distal end of Item # 3 creating a straight working channel beyond the end of Item # 3

[0124] In a still further embodiment, FIGS. 8 through 12 illustrate a windowed cannula kit or system 210 of the present invention. Referring first to FIG. 8, system 210 includes the flexible probe 218 having a treatment device 220 disposed on the distal end, a windowed needle 212 having a window 222 disposed at the side of the distal end of the needle, a reverse ramp window stylet 214 and flexible channeling stylet 216. As shown in FIG. 9, the reverse ramp window stylet 214 is placed inside the windowed needle 212 (shown as an 11G needle). The stylet 214 maintains the needle's column strength as well as preventing bodily fluids from entering the needle when the needle is advanced through tissue and into the cortical shell of bone.

[0125] For purposes of illustration, the treatment device 220 is shown as an RF probe. However, a number of other treatment devices may be used for treatment, e.g., including but not limited to ultrasound, radiation, fluidic heating through heat or stem, injection of bone cement or other compound/agent, light or laser energy, etc.

[0126] The treatment device 220 may ablate the treatment location, or stimulate it with a lesser amount of heat/energy or other treatment modality.

[0127] In addition to a treatment device, the methods and systems described below may also be used to deliver a diagnostic device (e.g. imaging system or the like) prior to or in conjunction with treatment.

[0128] Once in position, the reverse ramp window stylet 214 is removed and replaced with the flexible channeling stylet 216. As shown in FIG. 10, the channeling stylet 216 is advanced down toward the distal end of the windowed needle 212 to deflect out the side window 222 at an angle and channels through the bone to a predetermined distance. The channeling stylet 216 is preferably made from PEEK, which allows it to create a linear channel in the bone.

[0129] The channeling stylet 216 is then removed and, as shown in FIG. 11, the flexible functional

element **218** (with RF probe **220** at distal tip) is placed down the needle **212** and exits the side window **222** into the channel created by the flexible channeling stylet **216**. Because this system uses the flexible channeling stylet **216** to create a linear channel in the bone, the functional element **218** is delivered accurately and predictably within the desired treatment location within the vertebral body (e.g., intraosseous nerve target such as the BVN).

[0130] FIG. **12** illustrates a detailed view of the distal end of the windowed needle **212** and window **222**. The distal end of the window **222** comprises a concave ramp **224** to support deployment of the channeling stylet **216**. Distal to the window **222** is a beveled leading edge **226**. The needle **212** shown in FIG. **12** is comprised of stainless steel and has a 7 mm window.

[0131] Although the description above contains many details, these should not be construed as limiting the scope of the invention but as merely providing illustrations of some of the presently preferred embodiments of this invention. Therefore, it will be appreciated that the scope of the present invention fully encompasses other embodiments which may become obvious to those skilled in the art, and that the scope of the present invention is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” All structural, chemical, and functional equivalents to the elements of the above-described preferred embodiment that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Moreover, it is not necessary for a device or method to address each and every problem sought to be solved by the present invention, for it to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112, sixth paragraph, unless the element is expressly recited using the phrase “means for.”

Claims

1. A method of ablating a basivertebral nerve in a vertebral body of a patient, the method comprising: providing an introducer and an instrument, the introducer having a shaft defining a central channel and an opening at a distal tip of the introducer, the instrument comprising a proximal handle and a distal deflectable section; inserting the introducer through a pedicle of the vertebral body toward a cancellous portion of the vertebral body; delivering the instrument into the central channel of the introducer and toward the cancellous portion of the vertebral body; advancing the distal deflectable section of the instrument past the opening at the distal tip of the introducer and along a curved path in the cancellous portion of the vertebral body; and applying bipolar radiofrequency energy between a first electrode and a second electrode of a bipolar RF probe disposed in the curved path in the immediate vicinity of the basivertebral nerve and proximal to a junction of the basivertebral nerve to ablate at least a portion of the basivertebral nerve.
2. The method of claim 1, wherein the instrument further comprises a locking member at the proximal handle of the instrument.
3. The method of claim 2, wherein the locking member is in a locked position as the distal deflectable section of the instrument is advanced relative to the opening at the distal tip of the introducer.
4. The method of claim 1, wherein inserting the introducer comprises inserting a stylet into the central channel of the introducer such that the stylet extends beyond the opening at the distal tip of the introducer and advancing the introducer and stylet simultaneously such that the stylet pierces bone as the introducer is delivered through the pedicle of the vertebral body.
5. The method of claim 1, wherein the instrument comprises a cannula and a stylet.
6. A method of ablating a basivertebral nerve in a vertebral body of a patient, the method

comprising: inserting an introducer through a pedicle of the vertebral body toward a cancellous portion of the vertebral body, the introducer having a shaft defining a central channel and an opening at a distal tip of the introducer; delivering an instrument into the central channel of the introducer, the instrument comprising a proximal handle and a distal deflectable section; advancing the distal deflectable section of the instrument past the opening at the distal tip of the introducer and along a nonlinear path; and delivering bipolar radiofrequency energy from a first electrode of a bipolar RF probe to a second electrode of the bipolar RF probe located in at least a portion of the nonlinear path in the immediate vicinity of the basivertebral nerve and proximal of a junction of the basivertebral nerve to ablate at least a portion of the basivertebral nerve.

7. The method of claim 6, wherein the distal deflectable section of the instrument curves at a radius of no more than 1.0 inch relative to the longitudinal axis of the introducer when the distal deflectable section is positioned proximal to the junction of the basivertebral nerve.

8. The method of claim 6, wherein the instrument further comprises a locking member located at the proximal handle.

9. The method of claim 6, wherein the distal deflectable section of the instrument comprises a plurality of reliefs or cuts configured to facilitate bending of the distal deflectable section of the instrument.

10. The method of claim 6, wherein the treatment probe contributes to fluidic heating near the target area.

11. The method of claim 6, wherein the vertebral body is a lumbar vertebral body.

12. The method of claim 6, wherein the vertebral body is an S1 vertebral body.

13. The method of claim 6, further comprising injecting a compound or agent in the immediate vicinity of the basivertebral nerve.

14. A method of ablating a basivertebral nerve in a vertebral body of a patient, the method comprising: inserting an introducer through a pedicle of the vertebral body toward a cancellous portion of the vertebral body, the introducer having a shaft defining a central channel and an opening at a distal tip of the introducer, wherein inserting the introducer comprises inserting a stylet into the central channel of the introducer such that the stylet extends beyond the opening at the distal tip of the introducer and advancing the introducer and stylet simultaneously such that the stylet pierces bone as the introducer is delivered through the pedicle of the vertebral body; delivering an instrument through the central channel of the introducer, the instrument comprising a proximal handle and a distal deflectable section; advancing the distal deflectable section of the instrument past the opening at the distal tip of the introducer to form a curved path in the cancellous portion of the vertebral body; advancing a bipolar RF probe into the curved path such that a first electrode and a second electrode of the bipolar RF probe are in the immediate vicinity of the basivertebral nerve and proximal of a junction of the basivertebral nerve; applying bipolar radiofrequency energy between the first electrode and the second electrode to ablate at least a portion of the basivertebral nerve.

15. The method of claim 14, further comprising locking a locking member disposed at the proximal handle of the instrument.

16. The method of claim 14, wherein the distal deflectable section of the instrument comprises a plurality of reliefs or cuts.

17. A method of ablating a basivertebral nerve in a vertebral body of a patient, the method comprising: providing an introducer and one or more instruments, the introducer comprising a shaft defining a central channel and an opening at a distal tip of the introducer, the one or more instruments comprising a proximal handle and a distal deflectable section; inserting the introducer through a pedicle of the vertebral body toward a cancellous portion of the vertebral body; delivering the one or more instruments through the central channel of the introducer; advancing the distal deflectable section of the one or more instruments past the opening at the distal tip of the introducer into the cancellous portion of the vertebral body along a curved path; positioning the

distal deflectable section proximal to a junction of the basivertebral nerve; advancing a bipolar RF probe past the opening at the distal tip of the introducer such that a first electrode and a second electrode of the bipolar RF probe are positioned in the curved path in the immediate vicinity of the basivertebral nerve; and delivering energy from the first electrode to the second electrode to ablate at least a portion of the basivertebral nerve.

18. The method of claim 17, wherein, after the instrument is advanced relative to the introducer, the distal deflectable section of the one or more instruments curves at a radius of 0.6 inches to 1.0 inches and arcs to an angle of 5 degrees to 90 degrees.

19. The method of claim 18, wherein the angle of 5 degrees to 90 degrees is relative to a longitudinal axis of the introducer.

20. The method of claim 17, wherein the one or more instruments comprises a deflectable cannula and a deflectable stylet.
